A JOINTLY SURJECTIVE FAMILY OF FLAT MORPHISMS OF FINITE PRESENTATION COVERS EVERY OBJECT AFTER AN FPPF BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The criterion for a single flat morphism of finite presentation to admit fppf local lifts of objects. Invoked at the start of the counted proof (source lines 5925-5927) as a pointer to a lemma of this chapter. Provides the covering for the morphism from the coproduct of the family.
- The description of a morphism from a scheme into a disjoint union of stacks as a disjoint union decomposition of the scheme. Used at the end of the counted proof (source lines 5929-5933). Turns the lift to the coproduct into lifts indexed by the pieces of the decomposition.

Source reference. Original lemma lemma-surjective-flat-locally-finite-presentation: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5926–5926. Navigation pointer only: the counted text refers to the Stacks item lemma-surjective-flat-locally-finite-presentation, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f_j : \mathcal{X}_j \to \mathcal{X}$, $j \in J$ be a family of morphisms of algebraic stacks which are each flat and locally of finite presentation and which are jointly surjective, i.e., $|\mathcal{X}| = \bigcup |f_j|(|\mathcal{X}_j|)$. Then for every scheme U and object x of $\mathcal{X}$ over U there exists an fppf covering $\{U_i \to U\}_{i \in I}$, a map $a : I \to J$, and objects x_i of $\mathcal{X}_{a(i)}$ over U_i such that $f_{a(i)}(x_i) \cong y|_{U_i}$ in $\mathcal{X}_{U_i}$.*

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Proof. Apply Lemma lemma-surjective-flat-locally-finite-presentation to the morphism $\coprod_{j \in J} \mathcal{X}_j \to \mathcal{X}$. (There is a slight set theoretic issue here – due to our setup of things – which we ignore.) To finish, note that a morphism $x_i : U_i \to \coprod_{j \in J} \mathcal{X}_j$ is given by a disjoint union decomposition $U_i = \coprod U_{i,j}$ and morphisms $U_{i,j} \to \mathcal{X}_j$. Then the fppf covering $\{U_{i,j} \to U\}$ and the morphisms $U_{i,j} \to \mathcal{X}_j$ do the job. $\square$